

5 FEB 2025

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CE/FEB 2025/MEC331

**UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION**

COURSE : MACHINE ELEMENT DESIGN
COURSE CODE : MEC331
EXAMINATION : FEBRUARY 2025
TIME : 3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of four (4) questions.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
 - i) the Question Paper
 - ii) an Answer Booklet – provided by the Faculty
 - iii) a Formulas and Table Booklet – provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages

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QUESTION 1

- a) A screw jack with the ACME thread of mean diameter 30 mm, is used to lift the load of 6 kN. The screw jack has an efficiency of 40 % when lifting the load. The coefficient of thread friction is estimated as 0.15 and the collar friction is neglected. The thread angle normal to the plane and helix angle is 14.28° and 10.31° respectively.

- List **THREE (3)** factors that contribute to the efficiency of ACME thread power screw.
- If the coefficient of thread friction used is 0.2, then specify the existing of self-locking screw.
- Calculate the efficiency of screw jack when the screw is changed to the square thread. [CO1/PO1/C1/DK3/DP1/DP2/DP4]

(8 marks)

- b) A pressurized cylinder shown in **Figure Q1(b)** is used to hold pressurized gas at external force of 46 kN. The cylinder is joined to cylinder head by 6 bolted joints and reused connection. The bolt to be used is coarse thread with class number 9.8 and the safety factor 2.6. All dimensions are in millimeters. The modulus elasticity of bolt is 2 times the modulus elasticity of the part and the effective area of the part is equal to 3 times of the bolt area.

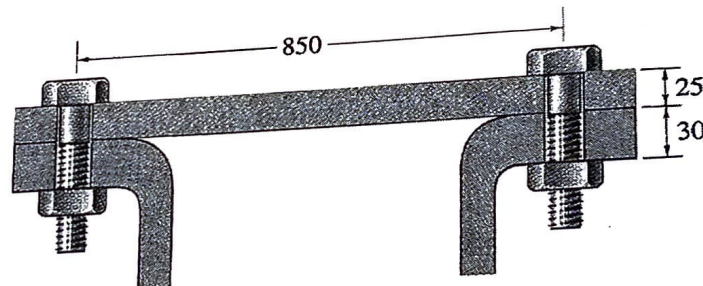


Figure Q1(b)

- Determine the joint stiffness factor and the size of the bolt.
- If the number of bolts are reduced to 4, state whether the plate will be separated. Prove your answer by calculation.
- Calculate the safety factor if the metric grade of the bolt is 4.6 then explain the strength of pressure cylinder. [CO3/P03/C5/DK5/DP2/DP3/DP4/DP5]

(16 marks)

QUESTION 2

- a) Cylindrical roller bearings 03 series are the most common type of roller bearing. They are commonly used in electric motors. The electric motor is intended for a load rating of 102100 N operation at 900 rpm.

- State **TWO (2)** characteristics of cylindrical roller bearing.
- Determine the size of the bearing.
- Due to improvement, the bearing is changed to angular contact ball bearing. State characteristic of the bearing. [CO1/PO1/C1/DK3/DP1/DP2/DP5]

(8 marks)

- b) A compressor is driven by an electric motor with a flat leather belt. The electric motor pulley is 300 mm diameter and rotates at 2400 rpm. The compressor pulley has a diameter of 500 mm and is located at a distance of 576 mm center to center from the electric motor pulley. The coefficient of friction is 0.25 and the density of the belt is 1000 kg/m³. The maximum permissible stress of the belt is 2.8 MPa and the loose side tension is 250 N.

- Calculate the cross sectional area of the flat belt and power transmitted.
- Conclude the effect of increasing the power transferred if the belt is changed to a round leather belt. This belt has the same cross sectional area as the answer in (i).
- Determine the power transmitted when the angle of contact is increased to 190° by using idler pulley and comment on your answer. [CO3/P03/C5/DK3/DK4/DK5/DP2/DP3/DP4]

(16 marks)

QUESTION 3

- a) The schematic drawing of a shaft is shown in **Figure Q3(a)**.

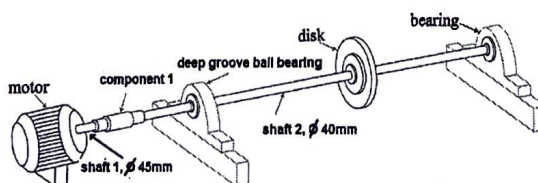


Figure Q3(a)

- Name component 1.
- State **TWO (2)** advantages if the bearing is changed to the needle bearing.
- A flat key is used on the shaft to attach the hub of rotating disk. Estimate the dimensions of the flat key. [CO1/PO1/C1/DK3/DP1/DP2/DP4]

(6 marks)

- b) The shaft shown in **Figure Q3(b)** is driven by pulley A from an electric motor and pulley B is connected to a compressor. The diameters of pulley A and B are 200 mm and 440 mm respectively. Three materials of the shaft used in this design are as shown in **Table Q3(b)** and the factor of safety is 2.

Table Q3(b)

Material	Yield Strength (MPa)
A	280
B	520
C	830

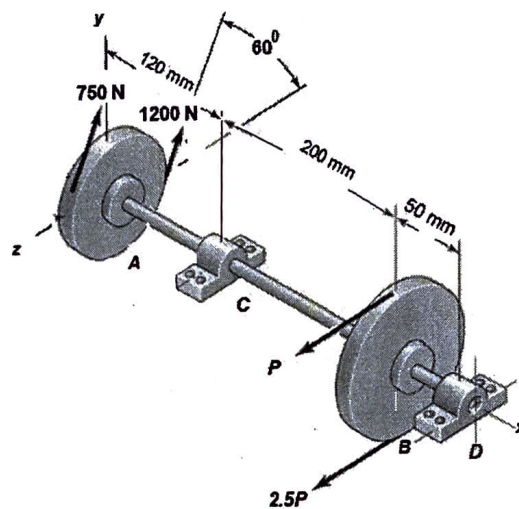


Figure Q3(b)

- Determine the belt tensions on pulley B.
- By applying the maximum energy distortion theory of failure, suggest the shaft diameter if material B is used.
- The customer wants to use material C due to the high yield strength. As the technical leader, explain to the customer how material C differs in terms of the size and cost when compared to material B. [CO2/P02/C4/DK3/DK4/DP1/DP3/DP6]

(20 marks)

QUESTION 4

- a) The spur gear is a simple gear used in washing machines.

- List **TWO (2)** advantages of spur gear.
- State how to increase the rotation of output gear speed when two gears are in mesh.
- State what will probably happen to the washing machine if the spur gear is replaced with the helical gear. [CO1/P01/C1/DK3/DP1/DP2/DP4]

(6 marks)

- b) The gear train in **Figure Q4(d)** is required to transmit the power of pinion 1. The number of teeth on gears 1, 2, 3 and 4 are 50, 125, 25 and 75 respectively. The module of the gear is 4 mm with the pressure angle of 20° .

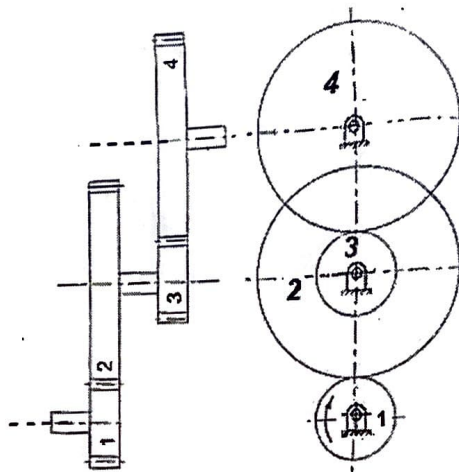


Figure Q4(d)

- Determine the speed ratio from gear 4 to pinion 1.
- Calculate the tangential and radial forces between gears 1 and 2 and between gears 3 and 4 if the power transmission is 40 kW at 1500 rpm of pinion 1.
- Find the new speed ratio if the positions of gear 3 and gear 4 are switched between them. [CO2/P02/C4/DK3/DK4/DP1/DP2/DP3]

(20 marks)

END OF QUESTION PAPER